

KAMENSKOYE, Russian Federation

WMO: 257450

Lat: 62.467N

Lon: 166.213E

Elev: 40

StdP: 100.84

Time Zone: 12.00 (E12)

Period: 90-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-41.4	-39.6	-44.0	0.1	-41.1	-42.1	0.1	-39.4	22.6	-18.5	20.3	-17.3	1.2	100	0.839

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.0	22.6	14.8	20.9	14.2	19.0	13.4	15.9	20.7	14.9	19.3	14.0	17.7	3.6	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.9	9.9	17.3	13.0	9.4	15.8	12.3	8.9	15.2	44.9	20.8	42.0	19.3	39.6	17.8	19.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
13.1	10.6	9.2	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-5.6	-23.7	-21.8	-19.0	-8.6	2.4	10.8	14.3	11.9	5.7	-4.0	-14.2	-21.6
	DBStd	15.36	9.65	9.39	8.05	6.58	3.94	3.05	2.45	2.92	3.69	6.38	8.64	9.88
	HDD10.0	5947	1046	889	900	559	236	26	1	15	135	435	727	979
	HDD18.3	8727	1304	1123	1158	809	494	225	126	201	380	693	977	1237
	CDD10.0	264	0	0	0	0	1	51	135	73	5	0	0	0
	CDD18.3	2	0	0	0	0	0	0	2	1	0	0	0	0
	CDH23.3	39	0	0	0	0	0	6	27	7	0	0	0	0
	CDH26.7	4	0	0	0	0	0	1	1	3	0	0	0	0
Wind	WSAvg	3.7	4.4	4.6	4.1	3.8	3.7	3.8	3.4	3.2	2.8	3.4	4.1	3.9
Precipitation	PrecAvg	412	36	30	24	20	17	30	49	56	43	31	34	31
	PrecMax	725	120	179	73	59	44	97	127	98	141	78	152	102
	PrecMin	248	1	0	1	2	1	0	0	10	1	8	6	1
	PrecStd	109	28	31	18	14	11	21	30	24	28	15	24	25
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.3	0.2	2.7	5.3	16.3	23.7	25.9	22.7	17.1	8.2	3.4	3.0
		MCWB	0.4	-1.3	0.8	2.4	9.4	13.7	16.1	15.4	12.7	6.1	1.7	1.5
	2%	DB	-1.6	-2.7	-1.3	2.9	12.7	21.2	23.8	20.4	14.3	7.0	1.6	-1.6
		MCWB	-3.0	-3.5	-2.9	0.7	6.9	13.0	15.4	14.7	10.4	5.1	0.5	-2.5
	5%	DB	-6.6	-5.0	-3.9	1.8	10.4	19.2	21.9	18.4	12.7	5.1	-0.2	-4.8
		MCWB	-7.3	-5.5	-4.8	0.2	5.7	12.2	14.9	13.7	9.8	3.5	-1.1	-5.3
	10%	DB	-10.7	-8.0	-7.3	0.5	8.3	17.4	20.1	16.8	11.3	3.5	-2.9	-7.8
		MCWB	-11.3	-8.3	-8.1	-0.9	4.6	11.3	14.3	13.1	8.9	2.0	-3.7	-8.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.6	-1.1	0.7	2.8	9.4	14.5	17.3	17.2	12.9	6.6	2.1	1.4
		MCDB	2.2	-0.1	2.3	5.3	16.0	21.6	23.6	21.1	15.7	7.7	3.0	3.0
	2%	WB	-3.1	-3.5	-2.8	1.3	7.3	13.5	16.3	15.6	11.5	5.2	0.7	-2.3
		MCDB	-1.6	-2.8	-1.4	2.7	12.1	20.0	21.7	18.9	13.2	6.8	1.6	-1.6
	5%	WB	-7.4	-5.5	-4.8	0.2	5.9	12.7	15.5	14.4	10.3	3.7	-1.3	-5.3
		MCDB	-6.7	-5.0	-3.9	1.7	10.0	18.5	20.5	17.3	12.1	5.0	-0.2	-4.8
	10%	WB	-11.3	-8.3	-8.1	-0.9	4.8	11.7	14.7	13.5	9.3	2.2	-3.7	-8.2
		MCDB	-10.7	-7.9	-7.3	0.5	7.9	16.7	19.1	15.8	11.1	3.1	-2.9	-7.8
Mean Daily Temperature Range	5% DB	MDBR	6.1	6.7	7.7	7.7	6.9	9.3	9.0	7.8	7.5	5.6	6.4	6.2
		MCDBR	8.4	6.3	6.6	5.8	10.3	13.5	13.5	11.1	9.4	5.2	7.4	7.6
		MCWBR	7.6	5.7	5.7	4.4	5.6	6.5	6.5	5.8	6.0	3.8	6.8	7.1
	5% WB	MCDBR	8.4	6.3	6.7	5.6	9.8	11.9	11.7	8.4	7.4	4.4	7.2	7.6
		MCWBR	7.7	5.8	5.7	4.3	5.4	6.2	6.1	4.9	5.1	3.4	6.6	7.1
Clear-Sky Solar Irradiance	taub	0.197	0.257	0.317	0.384	0.452	0.414	0.498	0.476	0.337	0.293	0.222	0.181	
	taud	2.021	2.132	2.027	1.874	1.852	2.215	1.971	1.993	2.427	2.264	2.004	1.922	
	Ebn at Noon	491	698	763	776	752	806	711	685	766	644	415	286	
	Edh at Noon	33	66	109	158	177	126	157	141	75	58	33	19	
All-Sky Solar Radiation	RadAvg	0.28	1.12	2.86	4.97	5.15	5.25	4.98	3.58	2.30	1.13	0.38	0.12	
	RadStd	0.04	0.12	0.22	0.24	0.60	0.59	0.49	0.40	0.22	0.15	0.06	0.01	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page